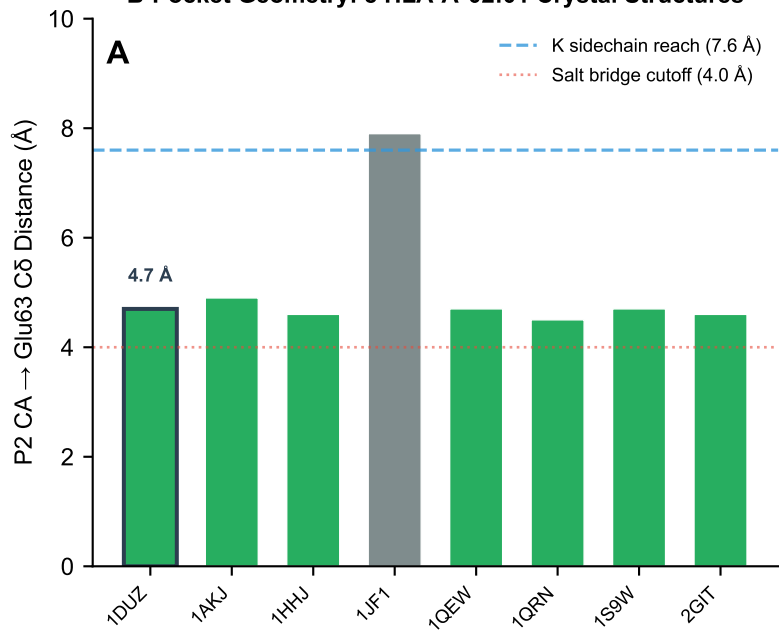
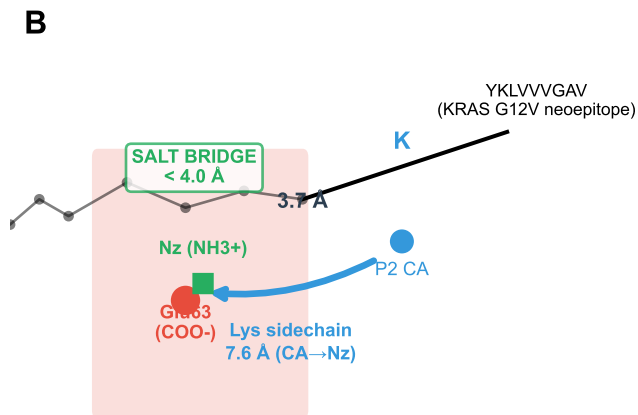


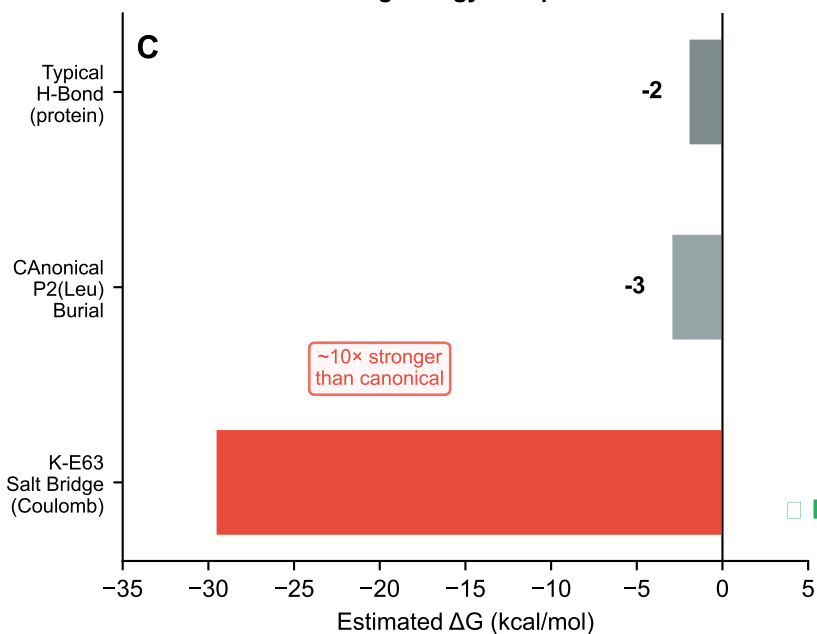
B-Pocket Geometry: 8 HLA-A*02:01 Crystal Structures



K-E63 Salt Bridge: Geometry Schematic



Binding Energy Comparison



K-E63 Salt Bridge: Key Metrics

D

P2 CA → Glu63 OE2	3.7 Å	(minimised 1DUZ)
K sidechain length (CA→Nz)	7.6 Å	(4 methylene + NH3+)
K reach margin	+3.9 Å	(sidechain overshoots E63)
Est. Nz → Glu63 OE	< 1 Å	(within salt bridge range)
Coulomb ΔE ($\epsilon=4$, $r=2.8\text{\AA}$)	-29.6 kcal/mol	(cf. Leu burial -3 kcal/mol)
B-pocket conservation (8 PDBs)	11.4 ± 0.1 Å	(avg CA-CA, stdev < 0.1)

E63→P2 CA (8 structures) 5.1 ± 1.1 Å (4.7 ± 0.2 excl. outlier)

□ K-E63 SALT BRIDGE: STRUCTURALLY & ENERGETICALLY FAVORABLE

First charged P2 in HLA-A*02:01? YES (no K at P2 in PDB)